THE Open GROUP

TOGAF® Standard – Architecture Deliverables Phase D

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Barangay San Antonio Health Department Enterprise Architecture

Barangay San Antonio Health Center, Pasig City

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1. Architectural Artifacts

While the Enterprise Metamodel is used to support the structuring of architectural information, most stakeholders do not need or wish to know the detail contained within the Architecture Content Framework in this manner. Therefore, the use of catalogs, matrices, and diagrams is introduced to facilitate the presentation of architectural information so that it may be used for reference and governance purposes more readily.

Catalogs are lists of building blocks of a specific type or related types, matrices are grids that show relationships between two or more entities, and diagrams are graphical renderings of architectural content.

In summary, the result of an Architecture Development Method (ADM) developed architecture consists of a number of defined Architecture Building Blocks populated into architecture catalogs, with relationships specified between those building blocks in architecture matrices, and/or also presented as diagrams, as appropriate to satisfy the stakeholder concerns.

The TOGAF® Standard provides a set of recommended artifacts, summarized as follows:

ADM Phase	**Artifact**
Preliminary Phase	Principles catalog
Phase A	Stakeholder Map matrix Value Chain diagram Solution Concept diagram Business Model diagram Business Capability map Value Stream map

ADM Phase	Artifact
Phase B	Organization/Actor catalog Driver/Goal/Objective catalog Role catalog Business Service/Function catalog Location catalog Process/Event/Control/Product catalog Contract/Measure catalog Business Capabilities catalog Value Stream catalog Value Stream Stages catalog Business Interaction matrix Actor/Role matrix Value Stream/Capability matrix Strategy/Capability matrix Capability/Organization matrix Business Footprint diagram Business Service/Information diagram Functional Decomposition diagram Product Lifecycle diagram Goal/Objective/Service diagram Business Use-Case diagram Organization Decomposition diagram Process Flow diagram Event diagram Business Capability map (updated) Value Stream map (updated) Organization map
Phase C: Data Architecture	Data Entity/Data Component catalog Data Entity/Business Function matrix Application/Data matrix Conceptual Data Diagram Logical Data Diagram Data Dissemination diagram Data Security diagram Data Migration diagram Data Lifecycle diagram
Phase C: Application Architecture	Application Portfolio catalog Interface catalog Application/Organization matrix Role/Application matrix Application/Function matrix Application Interaction matrix Application Communication diagram Application and User Location diagram Application Use-Case diagram Enterprise Manageability diagram Process/Application Realization diagram Software Engineering diagram Application Migration diagram Software Distribution diagram

ADM Phase	Artifact
Phase D	Technology Standards catalog Technology Portfolio catalog Application/Technology matrix Environments and Locations diagram Platform Decomposition diagram Processing diagram Networked Computing/Hardware diagram Network and Communications diagram
Phase E	Project Context diagram Benefits diagram
Requirements Management	Requirements catalog

2. Phase D: Technology Architecture Artifacts

All core applications are cloud-hosted by the City, PhilHealth, DOH or NNC; the health center needs working devices and a reliable connection. City-issued equipment is audited by the City about every six months and maintained once a year; barangay equipment is checked by the barangay IT team every three months.

2.1 Technology Portfolio Catalog

Item	Qty	Owner	Use	Condition
Laptops	4	City	Clinical staff and encoder (EMR, PhilHealth, LMIS)	10+ years old; slow; replacement requested
Desktop workstations	2	Barangay	Shared / front desk	Old
Laptop	1	Barangay	Shared	Working
Laptop (BNS)	1	Barangay	Barangay Nutrition Scholar	Good
Printers	4 (2 City, 2 Barangay)	City / Barangay	Encoder; front desk	One from 2025; one faulty
Mobile phone	1	Health center	Front desk	Working
Internet line (PLDT)	1	Barangay (~₱3,000 a month)	Whole center	About 5 outages a month; slow at peak
Internet line (LGU)	1	City	Whole center	Weak or unpaid; not used
Generator	1	Barangay	Building	Available
Vaccine refrigerator	1	City / DOH	Storage	Temperature checked by custodian
Dental chair; autoclave sterilizer; UV/dry-heat cabinet	1 each	City (+ donations)	Dental clinic	Good; autoclave meets DOH accreditation
Weighing kit (OPT)	1	City / NNC	BNS	Incomplete

2.2 Technology Standards Catalog

Area	Standard or rule	Source
Software installation	Center cannot install software; the City approves and runs seminars for new programs	Mr. Reyes
System changes	Minor: City Health Officer; major: City Mayor	Dr. Bonilla
Accounts	Individual EMR accounts; passwords managed by IT	Mr. Reyes
Equipment audit	City about every 6 months; barangay every 3 months; City maintenance yearly	Kapitan; Ms. Abong
Clinical equipment	DOH sterilizer standard for accreditation	Dr. Magpali
Records	Data Privacy Act (RA 10173); DOH 7-year retention	Ms. Abong

2.3 Application/Technology Matrix

Application	Hosting	Device	Network
City EMR / dental eMED	Vendor cloud (City)	City laptops	PLDT line or phone hotspot
PhilHealth HCI / YAKAP	PhilHealth cloud	City laptops	Same
LMIS, ITIS, FHSIS	DOH / City cloud	City laptops	Same
e-OPT Plus	NNC	BNS laptop	Same; field use offline

Application	Hosting	Device	Network
Google Drive, Messenger, Viber, Facebook	Public cloud (personal accounts)	Laptops and personal phones	Same; mobile data

2.4 Environments and Locations Diagram

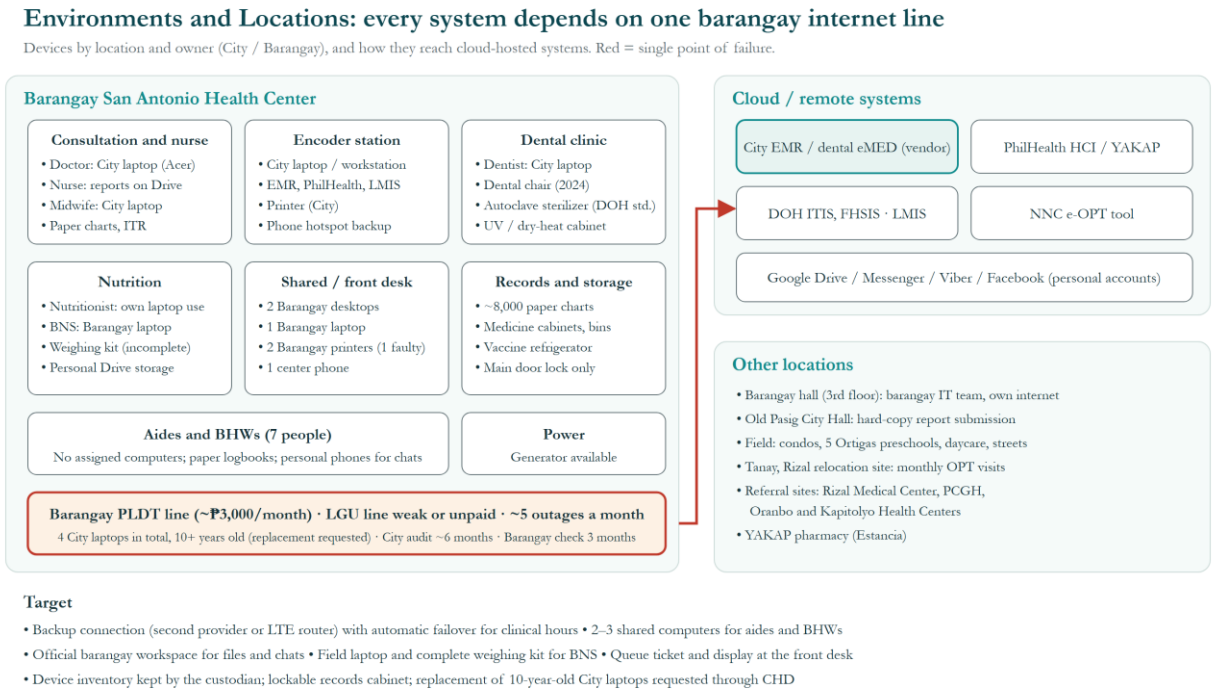


Figure 1. Environments and Locations Diagram.

How to read it: the large box is the health center, divided by area, with each device labelled City or barangay. Cloud systems are on the right; the network line is at the bottom. Red marks the single point of failure.

- **Devices:** 4 City laptops (doctor, midwife, dentist, encoder), all 10+ years old with replacement requested; barangay-owned: 2 desktops, 1 laptop, the BNS laptop and printers (one faulty).
- **Gap:** the 4 Pasig Health Aides and 3 BHWs have no assigned computer and use personal phones for work chats.
- **Network:** every cloud system is reached through one barangay-paid PLDT line (~₱3,000 a month); the LGU line is weak or unpaid; about 5 outages a month, covered by a phone hotspot.
- **Physical security:** the records room is locked only at the main door; a generator covers power cuts.
- **Targets:** backup internet (WP7), 2–3 shared computers (WP3), lockable records cabinet (WP4), City laptop replacement (WP8).

2.5 Network and Communications Diagram

All traffic from the health center to cloud systems passes through one barangay-paid PLDT line (about ₱3,000 a month); the LGU line is weak or unpaid, and staff use phone hotspots during outages (about 5 a month). The target adds a second provider or LTE router with automatic failover (WP7). See Figure 1.

2.6 Technology Gap Analysis

ID	Gap	Target	Controlled by
TG-1	One PLDT line (~₱3,000 a month); ~5 outages a month; slow at peak	Backup connection with failover	Barangay
TG-2	4 City laptops, 10+ years old; replacement requested	Replacement through CHD	City
TG-3	No computers for 4 aides and 3 BHWs	2–3 shared computers	Barangay
TG-4	No official storage or accounts	Barangay workspace with shared storage	Barangay
TG-5	No queue hardware	Ticket and display at front desk	Health center / Barangay
TG-6	BNS field data entered later from paper; weighing kit incomplete	Field laptop and complete weighing kit	Barangay
TG-7	Records room secured only at the main door	Lockable records cabinet; key log	Health center

A Abbreviations

Abbreviations, acronyms and reference codes used in this document and its figures, in alphabetical order.

Abbreviation	Meaning
ADM	Architecture Development Method, the step-by-step method of the TOGAF Standard
BHW	Barangay Health Worker
BNS	Barangay Nutrition Scholar
CHD	City Health Department (Pasig City)
DOH	Department of Health
e-OPT	Electronic Operation Timbang Plus tool of the National Nutrition Council
eMed / eMED	Name of the City's electronic medical record system; “dental eMED” is its separate dental system
EMR	Electronic Medical Record
FHSIS	Field Health Services Information System (DOH)
HCI	Health Care Institution (PhilHealth portal for health facilities)
Hon.	Honorable (title of elected officials)
IT	Information technology
ITIS	Integrated Tuberculosis Information System (DOH)
ITR	Individual Treatment Record (the patient's registration form)
LGU	Local Government Unit
LMIS	Logistics Management Information System
LTE	Long-Term Evolution (4G mobile internet)
NNC	National Nutrition Council
OPT / OPT Plus	Operation Timbang Plus: yearly weighing and measuring of children aged 0–59 months
PCGH	Pasig City General Hospital
PhilHealth	Philippine Health Insurance Corporation
PLDT	PLDT Inc., an internet and telephone provider
RA	Republic Act (a Philippine law)
TG-1 …	Technology gap reference number (Phase D)
TOGAF	The Open Group Architecture Framework (the TOGAF® Standard)
UA&P	University of Asia and the Pacific
UV	Ultraviolet
WP1 … WP9	Work package (a group of related changes delivered together)
YAKAP	PhilHealth Yaman ng Kalusugan Program (primary-care benefit)